
SUMMARY

- Machine Learning Engineer / Researcher (Ph.D.) with 7+ years building algorithms and **reproducible ML code** — from modeling to evaluation to delivery.
- Hands-on: Python (PyTorch, JAX, scikit-learn, NumPy), SQL; Git/Linux/Jupyter/pytest; experience with **time series**, **deep learning for PDEs (PINNs)**, and production-adjacent tooling (tests).
- Track record of **peer-reviewed publications** and collaborating across research and engineering teams.
- Practical use of LLMs to accelerate data processing (cleaning, summarization, light labeling) and research documentation; all outputs validated.

SKILLS

Programming	Python (PyTorch, JAX, scikit-learn, NumPy), MATLAB; SQL
ML	Deep learning, Physics-informed neural networks (PINNs), classification
Tools	Git, Linux, Jupyter, Bash, Slurm, pytest

EDUCATION

Ph.D. in Mathematics , University at Buffalo	2016.08 – 2021.08
Thesis: Numerical Solution to the Corner Angle Problem for an Elastic Solid with Anisotropic Surface Energy	
B.S. in Mathematics , Fudan University	2011.09 – 2016.06

PROJECTS

PINNs for High-Dimensional PDEs — University of Victoria	2023.11 – present
• Built a JAX-based neural solver for high-dimensional PDEs; broke long simulations into trainable steps (time-marching) and verified against numerical baselines. • Improved stability and accuracy by balancing loss terms and adjusting numeric precision; tracked residual and L2 error to convergence. • Wrote fast, tested training and evaluation components (JAX JIT).	
Railway Delay Prediction (Time Series) — Concordia / C&IE 2023	2022.09 – 2023.05
• Built an interpretable time-series (Markov) model to predict near-term delays on 1M rows; validated assumptions statistically and smoothed sparse transitions for stability. • Delivered Python code with clear metrics and documented evaluation.	
Baseball Outcome Modeling — Team project	2023.09 – 2023.10
• Cleaned/joined data (Retrosheet + 538 Elo); built season-aware simulations and logistic models; best model beat 538 Elo on Brier (≈ 0.2404 vs ≈ 0.2436).	
Side-Channel Analysis (Classification) — Security lab	2014.08 – 2016.06
• Applied PCA denoising and trace alignment; implemented template attacks (Python/MATLAB); small model changes improved key-recovery accuracy; recognized on the contest Hall of Fame.	

EXPERIENCE

Postdoctoral Fellow , University of Victoria	2023.11 – present
• Used PyTorch and JAX for ML research: benchmarks, documentation, and unit tests.	
Postdoctoral Fellow , Concordia University	2021.09 – 2023.05
• Designed algorithms and Python evaluation harnesses for graph/network problems (Age of Information); emphasis on metrics and reporting.	
Quant Analyst , NumerixS Quant	2023.06 – 2023.07
• Reduced false alerts in market-data integrity checks (kdb+/q) with a replay harness.	

SELECTED PUBLICATIONS

- *Approximation schemes for AoI minimization in UAV grid patrols*. Ad Hoc Networks, 2025.
- *A novel Markov model for near-term railway delay prediction*. C & IE, 2023.